

GREG PICKHARDT

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Professional Summary

Automation & Controls System Engineer with involvement transforming complex mechanical and electrical systems into **high-performance automated solutions**. Experienced in **developing PLC/HMI software** for precision industrial machinery and integration into fully automated production lines. Extensive hands-on experience in **Siemens TIA Portal and WinCC**. Adept at **bridging mechanical, electrical, and software engineering** to deliver scalable solutions for automated production environments.

Education

State University of New York at Delhi

- Bachelor in Mechatronics Technology | May 2023
 - Associate of Applied Science in Mechatronics Design | May 2021
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Technical Skills

Automation & Controls:

PLC/HMI programming, system integration, troubleshooting, control panel design, electrical schematics, pneumatics & hydraulics (P&ID).

Software:

Siemens TIA Portal, WinCC, LabVIEW, SolidWorks, SolidEdge, Autodesk Inventor, AutoCAD/AutoCAD Electrical, EPLAN, Hyper-V, VirtualBox, Microsoft Office.

Expertise & Coursework:

PLC Theory, Electrical and Control System Theory, Electrical Controls Troubleshooting, Controls System Design, Cybersecurity, Manufacturing and Material Processing, GD&T, ANSI/ISO Drafting Standards, Pneumatics & Hydraulics.

Professional Experience

NSH USA Corporation – Albany, NY

Automation & Controls System Engineer | Jul 2023 – Present

Machine tool manufacturer specialized in precision machine tools and automation systems primarily for the railway industry.

- **Software development and startup commissioning** for the Stanray® Wheel Truing Machine. Milling machine, capable of re-profiling railway wheels while directly under a locomotive.
- **Developed the WSM-450 (Wheel Set Metrology) machine from the ground up.** Automated inspection system with the capability to measure wheel set length features and runout. Has a measurement repeatability and accuracy of $\pm 0.002''$. Integration to handshake with other machines in a fully automated production line.
- **Improved diagnostics and fault recovery** on our machines.
- Collaborated with multidisciplinary teams to deliver **reliable automation solutions** tailored to rail and heavy industry requirements.

NSH USA Corporation – Albany, NY

Electro-Mechanical Engineering Intern | Jun 2022 – Dec 2022

- Developed a **LabVIEW program** to log gantry system coordinates and generate a **real-time 3D visual graph**, integrating data pulled directly from a **Siemens PLC database**.
- Machine startups for Stanray TN-84C wheel truing systems. Supporting **installation, commissioning, testing, and debugging** of PLC/HMI software and ensuring proper system function.
- Designed and documented wiring, schematics, and P&ID diagrams, ensuring **ISO** compliant systems.
- Designed an **OSHA** compliant structural staircase and operator platform for a Stanray M-Series pit, ensuring compliance with **safety and accessibility standards**.
- Performed quality control inspections using a Coordinate Measuring Machine (CMM) to verify part tolerances and ensure compliance with **GD&T** callouts.

Superior Ice Cream Equipment, LLC – Bow, NH

Mechanical Engineering Intern | May 2021 – Aug 2021

- Transposed mechanical sketches into **fully functional 3D SolidWorks assemblies**, accelerating prototyping and design validation.
- Redesigned parts and components, **enhancing machine reliability and reducing maintenance**.
- Built and wired control panels.